

LIFE SCIENCES • AWS-NATIVE • GOVERNED BY DESIGN

Healthcare & Life Sciences Agentic AI Suite

Eight bounded, regulator-aware agents for pharma, biotech, medtech & CROs — with a governed platform, an MCP authorization gateway on Amazon Bedrock AgentCore, and one-click AWS deployment.

Field enablement for sales, solution architects & delivery

From copilots that summarize to agents that get work done

Bounded, not autonomous

Agents monitor events, assemble evidence, plan, call approved tools, and draft regulated artifacts — humans make every medical, safety, quality and regulatory decision.

Regulators have a framework

FDA & EMA published joint good-AI principles (Jan 2026): define the model's context of use, apply credibility controls proportional to risk, keep humans accountable.

The work is ready for it

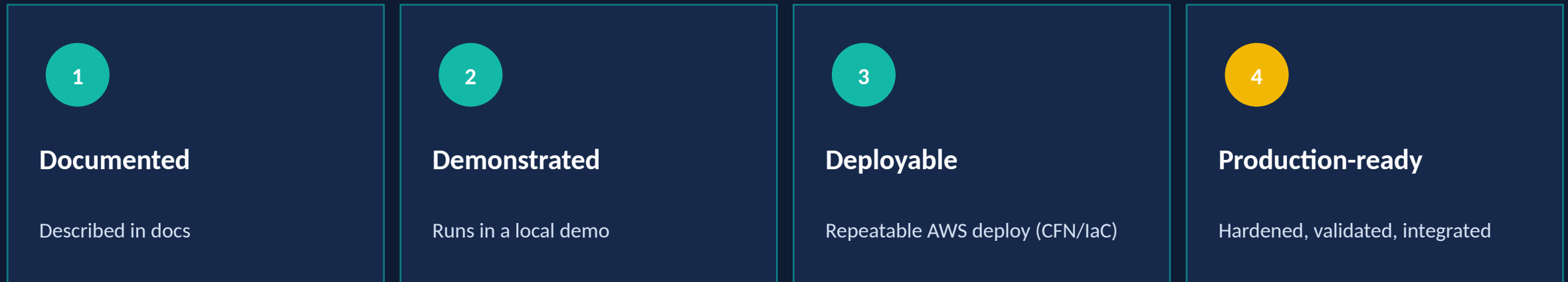
Regulatory writing, pharmacovigilance, clinical ops, quality/CAPA, RWE and medical affairs are high-volume, evidence-driven, and human-gated — ideal first agents.

Buyers want governed, not clever

The blocker to scaling agents in Life Sciences is trust: data residency, validation, auditability. Win there and the use cases follow.

The agents are not the product. The governed platform that makes them deployable is.

Anyone can demo a clever agent. Shipping one into a regulated Life Sciences environment requires identity, authorization, audit, validation and human control — the platform layer below.



This accelerator ships at Demonstrated + Deployable-by-design — honest about what is real today.

Eight Life Sciences agents

Each: bounded workflow • governed tool access • human-in-the-loop • audit trail • AWS-native deploy

01

Regulatory Writing

Drafts submission sections & responses;
grounding-checked

02

Pharmacovigilance

ICSR intake → MedDRA coding →
seriousness → narrative

03

Clinical Trial Ops

TMF completeness, data queries, study-
health briefing

04

Site & Patient Matching

Eligibility → de-identified cohort → site
ranking, equity

05

Quality / CAPA

Complaint triage, root cause, CAPA
drafting (QMSR)

06

Protocol Design

Guidance search, feasibility, protocol
section drafting

07

RWE / HEOR

Cohort definition, evidence synthesis,
data-quality flags

08

Medical Affairs / MSL

Compliant, cited pre-call briefs; MLR
submission

How every agent works — bounded autonomy



Autonomy is earned, not assumed: agents start at search & summarize and graduate to reversible, low-risk actions only after evidence. They never make the final medical, safety, quality or regulatory decision.

Six layers that make agents governable

1	User experience	Inside the apps people already use — RIM, safety DB, QMS, CRM, Teams
2	Supervisor + specialist agents	Decompose work; specialist agents per task; orchestrate
3	Tool & integration layer	The MCP authorization gateway — governed front door to every system
4	Data & semantic layer	Governed lake, vector search, ontologies (MedDRA/WHODrug), lineage
5	Models & deterministic services	LLM for drafting; validated stats/rules engines for calculation
6	Governance & observability	Versioning, audit, grounding, HITL, evals, kill-switch, monitoring

The MCP authorization gateway

Deployed as Amazon Bedrock AgentCore Gateway + AgentCore Identity. No agent touches a system of record directly.

1. Authenticate

Verified IdP claims; deny on missing identity

2. Authorize

Deny-by-default; agent grant $\cap$ user entitlement

3. Human approval

Required for any write / irreversible tool

4. Scoped token

Short-lived, single-tool; no standing credentials

5. Invoke

Through the validated connector to the system

6. Audit

PHI-masked, tamper-evident, with lineage

An agent can never exceed the human it acts for. Least privilege is enforced as an intersection — the same semantics in the Python reference and in AgentCore.

Governance & evaluation, not bolted on

● Grounding

Every number & entity in a regulated artifact must trace to source — catches hallucinated dose, endpoint, institution

● Prompt registry

Prompts are hash-pinned & version-controlled; CI fails on un-reviewed drift (model-risk discipline)

● Eval harness

Structural + grounding checks over golden artifacts run in CI — a prompt change can't silently degrade output

● HITL gates

High-risk actions are framework-enforced to require a verified human approval, not just documented

● Red team

Injected instructions can't change authorization; grounding catches smuggled facts; PHI masking blocks exfiltration

● Fairness

Flags demographic under-representation in proposed cohorts (FDA Diversity Action Plan posture)

One control envelope, many regulations

HIPAA / GDPR

PHI masking · in-account inference · de-identification · data residency

21 CFR Part 11

Tamper-evident audit trail · e-signature linkage at HITL gates · record integrity

GxP (GCP/GLP/GMP)

ALCOA+ data integrity · CSV-ready · prompt & model version control

GVP / ICH E2B(R3)

ICSR completeness checks · expedited-reporting logic · safety audit

FDA QMSR (Feb 2026)

CAPA traceability · complaint handling · ISO 13485 alignment

EU AI Act

Documented context of use · logging · human oversight · risk classification

Compliance comes from the whole system — technology, configuration, validation, procedure and evidence — not from any single service.

Quick-deploy, customer-isolated, in-account

Two paths to AWS

Container lift — AgentCore Runtime

Wrap the LangGraph agent in the AgentCore contract (/invocations, /ping, ARM64). Fastest; code unchanged.

Native rebuild — Strands + Step Functions

Deterministic core in Lambdas, Bedrock drafting, waitForTaskToken human gate. Highest managed fidelity.

Quick deploy — CloudFormation

Nested stacks: VPC, KMS + Guardrail, append-only audit, WORM store, AgentCore Gateway. Terraform parity included.

Customer-isolated data architecture

CloudFront + WAF → ALB (Cognito/Okta)

ECS Fargate — UI + agent worker

Bedrock AgentCore Gateway + Identity

Bedrock + Guardrails (in-account inference)

Aurora PostgreSQL — workflow state

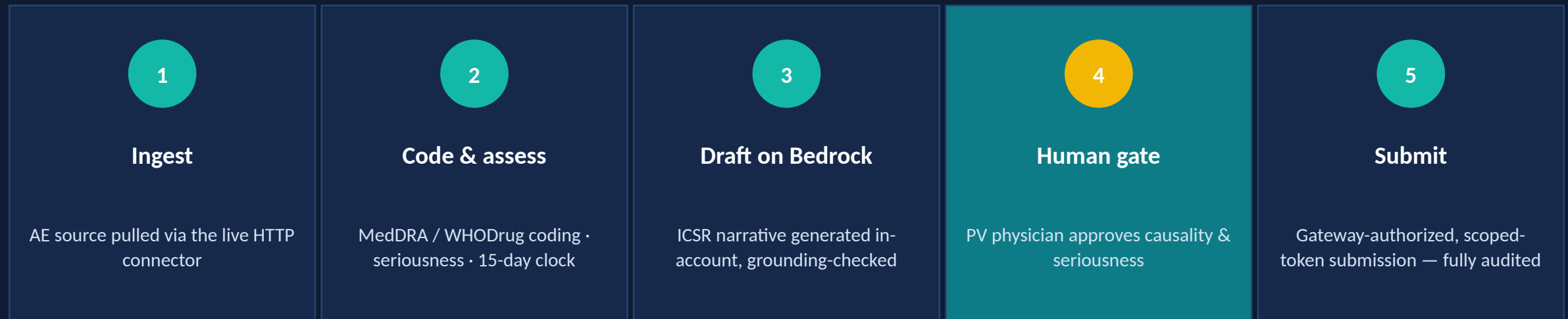
DynamoDB — append-only audit (Part 11)

S3 Object Lock — WORM regulated artifacts

KMS · Secrets Manager · CloudWatch

Pharmacovigilance, live on Bedrock + a real connector

One agent wired end-to-end: live Amazon Bedrock inference and a real HTTP safety-system connector — swap one URL from the local reference service to the customer's Argus / Veeva Safety gateway.



Runs today with no API key in demo mode; flip env vars for live Bedrock + live connector. `python demo/demo_live.py`

Where the value lands

75%

faster first draft of a benefit-risk summary (Agent 01)

~8 hrs

saved per ICSR case vs. manual processing (Agent 02)

Continuous

regulatory-intelligence & TMF monitoring vs. periodic, manual

Pre-review

grounding & compliance defects caught before a human reads the draft

Illustrative; the assessment instruments the customer's own baselines so the business case is their data, not a generic slide.

A staged path from proof to production



Plus: objection-handling, competitive positioning, ROI model, TPRM due-diligence packet, and stakeholder security briefings for CISO, CMO, Regulatory, PV, Quality, Privacy & Procurement.

Platform over point tools; partner over DIY

vs. building it yourselves

Internal teams underestimate identity, authorization, audit, validation and eval — the 80% that isn't the agent. We ship that as a reusable platform.

vs. vendor-embedded AI

Veeva/Argus/Medidata add AI inside their walls. This suite spans systems with one governed gateway, one audit standard, one control model.

vs. general-purpose platforms

Horizontal copilots aren't built for GxP, Part 11, GVP or QMSR. This is Life-Sciences-native, with the regulatory controls in the architecture.

A working, tested accelerator

8

agents + 8 AWS-native rebuilds

480+

tests passing across the suite

1

governed gateway = AgentCore
Gateway

CFN

one-click quick-deploy +
Terraform

Every agent runs offline with no API key (demo mode), flips to live Bedrock + live connectors by config, and ships with code, tests, governance evidence, runbooks and GTM material. *Production-readiness — hardening, full validation, customer integration — is the engagement.*

LET'S PICK THE FIRST WORKFLOW

Start with one measurable workflow

1. Pick a bounded, high-volume workflow (regulatory writing, PV intake, TMF, CAPA).
2. Run the assessment — context of use, risk, control map, the customer's own ROI baseline.
3. Stand up the demo on AWS; prove governance and human control end-to-end.
4. Scale by reusing the platform — identity, gateway, audit, evals — across agents.

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